



BSc (Hons) in Information Technology – Year 3

IT3060 – Human Computer Interaction

Lab Exercise 03

Design & Prototyping for the FitFlow App Redesign

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Activity 1: Review User Requirements

The three pain points given in the case study map directly onto the themes validated through the FitFlow research in Labs 1–2, and onto the requirements already derived in Lab 2, Activity 8. Restating them here against that evidence keeps the design phase grounded rather than starting from a blank page.

1.1 Pain Point → Evidence → User Need

Pain point (case study)	Linked research evidence (Lab 1–2)	Resulting user need
Limited personalization	Theme: “Lack of Personalization & AI Trust” (D-003, D-004, D-007, D-008) — 4 of 5 interviewees independently raised this.	See a short reason behind every AI recommendation and be able to override or adjust it (FR-01, FR-02, FR-08).
Cumbersome workout / meal logging	Theme: “Nutrition Tracking Issues” — “6 taps to log a snack” (D-001), 42% analytics drop-off (D-009).	Log a meal or exercise change in 3 taps or fewer, with recent items and barcode entry (FR-03, FR-04).
Poor progress visualization	Theme: “Poor Motivation” — no streak/re-engagement nudges (D-011); users disengage silently.	See simple, glanceable progress trends and get a nudge before a streak lapses (FR-06, FR-10).

1.2 Key User Needs for the New Features

- AI-powered workout plans: transparent (rationale shown), adjustable (manual override), and safe (respects injury/medical constraints — FR-01, FR-02, FR-08).
- Social / community features: motivating but privacy-safe — visibility must be set before joining any challenge (FR-05, FR-09, NFR-06).
- Nutrition tracking: fast entry from recent/favorite items or barcode, with a visible rationale for targets (FR-03, FR-04).

1.3 Prioritized Requirements (carried forward from Lab 2, Activity 8)

ID	Requirement	Priority
FR-01	Show a short rationale whenever an AI-generated recommendation is shown.	High
FR-02	Allow users to manually edit, swap, or override any part of an AI-generated plan.	High
FR-03	Allow logging a meal in three taps or fewer using recent/frequent/favorite items.	High
FR-05	Require a visibility level (private/friends/public) before joining a community challenge.	High
FR-07	Provide a guided onboarding walkthrough on first launch.	High
FR-08	Allow recording injury/medical constraints the AI must not violate.	High
FR-06	Send a reminder before an active streak lapses.	Medium
FR-10	Display progress analytics and trends over a selectable period.	Medium

1.4 Core User Flows

Flow	Steps
Home Dashboard	Open app → see today's AI plan + short rationale → see quick stats → see a community challenge preview → branch into Plan / Log / Community / Progress.
AI Workout Planner	See rationale banner → review exercise list → swap/override an exercise or log an injury constraint → start workout.
Nutrition Logger	Search or scan a meal or tap a recent/favorite item → item logged in ≤ 3 taps → see today's target with rationale.
Community Feed	Set visibility (private/friends/public) → browse active challenges → join → view leaderboard.
Progress Tracking	View monthly activity chart → view workout/duration/weight-trend stats → see a streak reminder if at risk of lapsing.

Activity 2: Crazy 8 Sketching

For each of three core screens — Home Dashboard, AI Workout Planner, and Progress Tracking (the additional screen chosen alongside the two required screens) eight rapid layout ideas were sketched in eight minutes, digitally, to force a wide spread of structural approaches before committing to any one direction. Sketches deliberately vary the position of the AI plan/rationale, the density of information, and how social/progress elements are surfaced.

Crazy 8s — Home Dashboard (8 sketches / 8 minutes)

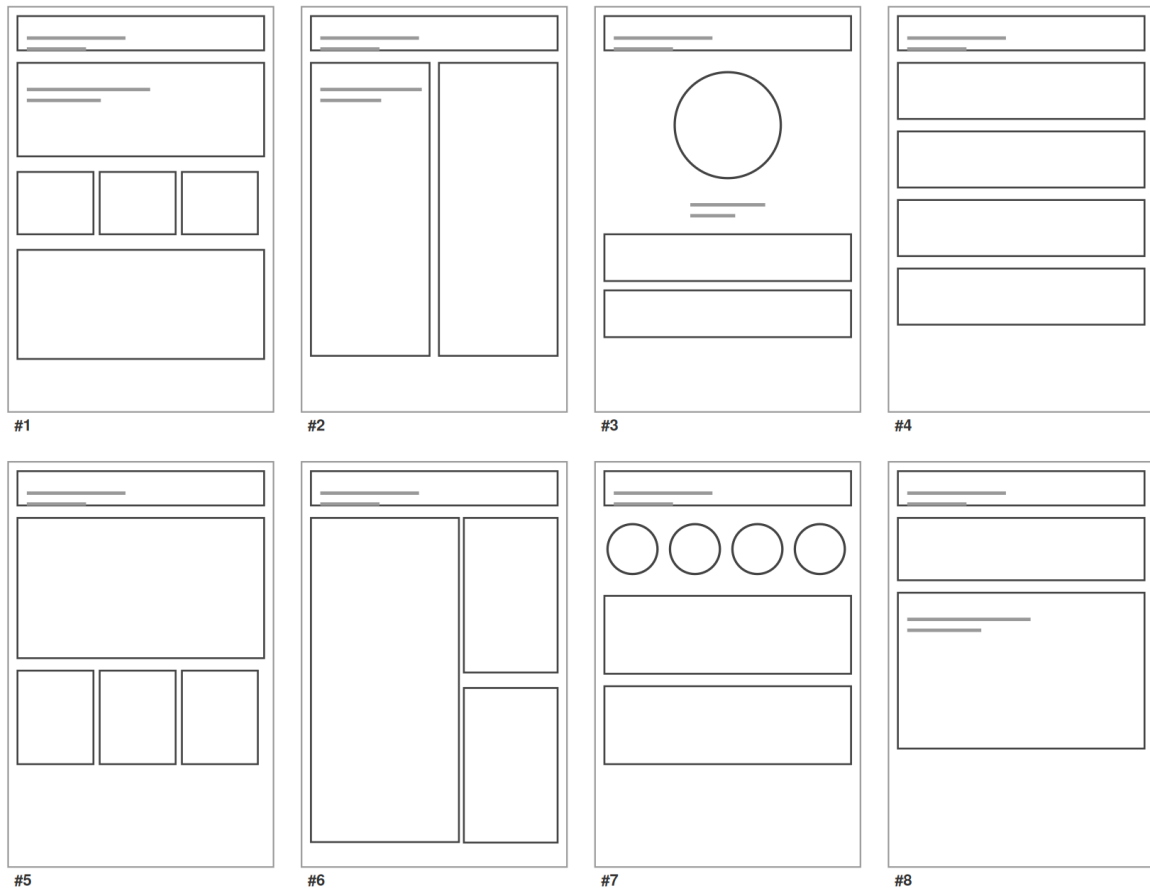


Figure 1: Crazy 8s — Home Dashboard (8 layout ideas in 8 minutes).

Crazy 8s — AI Workout Planner (8 sketches / 8 minutes)

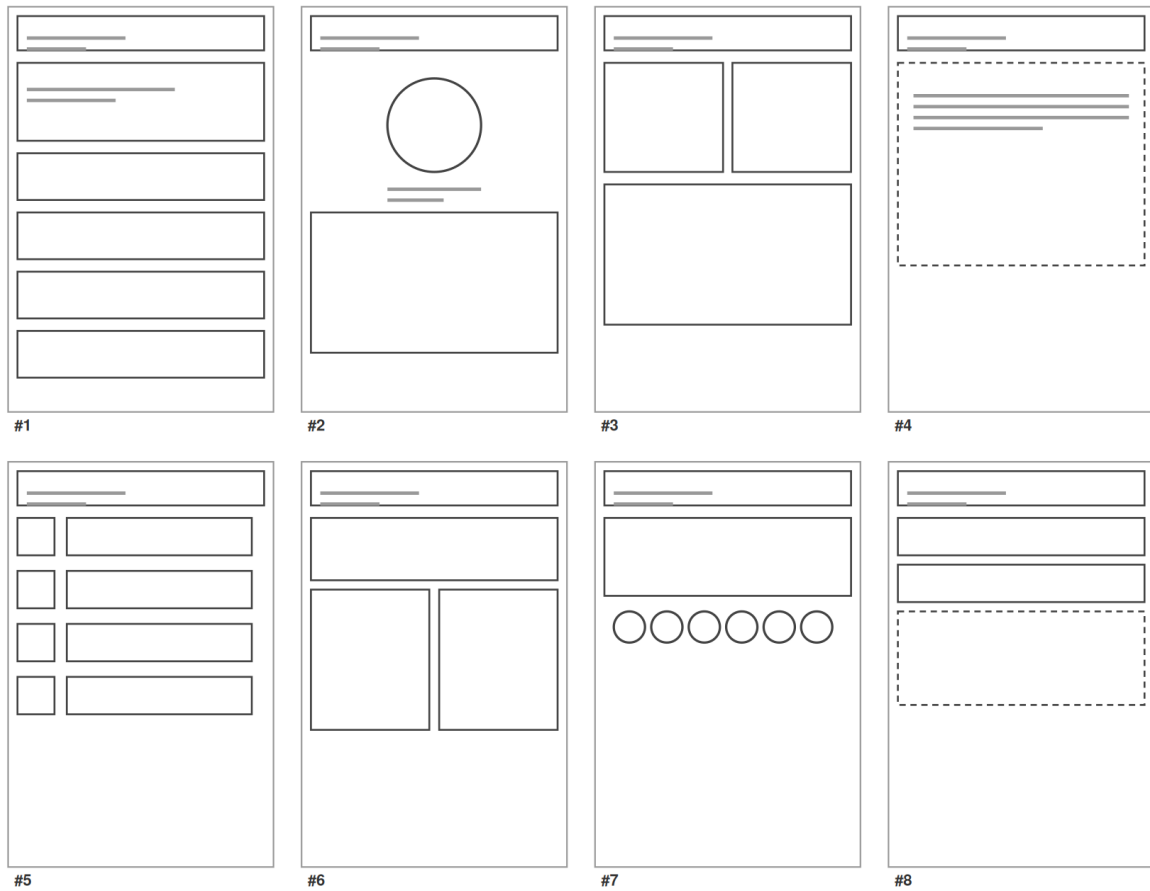


Figure 2: Crazy 8s — AI Workout Planner (8 layout ideas in 8 minutes).

Crazy 8s — Progress Tracking (8 sketches / 8 minutes)

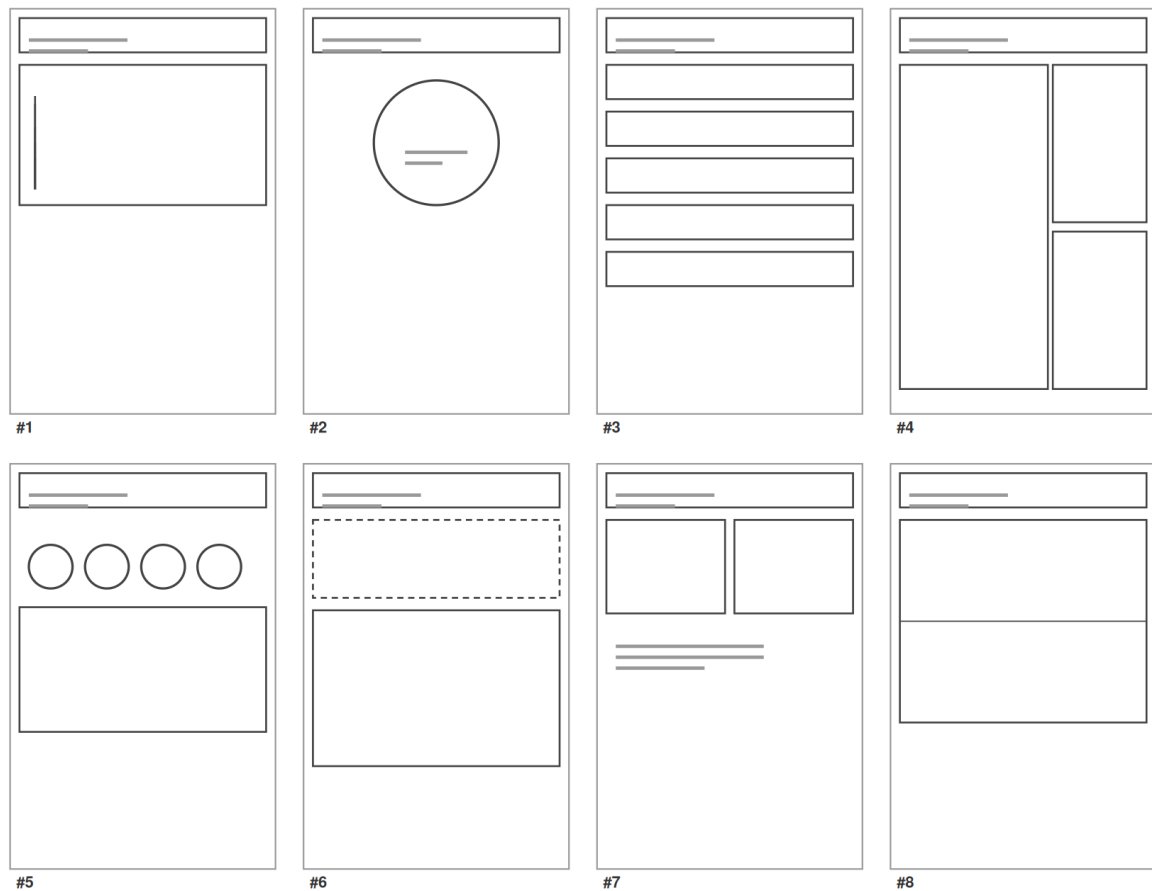


Figure 3: Crazy 8s — Progress Tracking (8 layout ideas in 8 minutes).

2.1 Promising Ideas Carried Forward

- Home #1 and #5 (rationale line directly under a hero AI-plan card) kept, because it makes FR-01 visible without an extra tap.
- Home #3 (progress-ring hero) kept as an alternative direction emphasizing glanceable progress (relevant to FR-10).
- Planner #1 (vertical list with a swap icon per row) kept, directly supports FR-02 (override) with minimal extra UI.
- Planner #2 (single-exercise carousel) kept as a lower-density alternative for quick review on the move.
- Progress #1 and #6 (bar chart + streak card grouped together) kept, groups FR-10 and FR-06 in one glance.

Activity 3: Produce Multiple Interface Variants

The strongest ideas from the Crazy 8 sheets were developed into three distinct, complete interface variants. Each variant applies one consistent layout philosophy across the app rather than mixing approaches screen-by-screen, so it can be judged as a coherent direction. Home Dashboard and AI Workout Planner are shown below as the representative pair of core screens; the same layout philosophy was applied consistently to Nutrition Logger, Community Feed, and Progress Tracking within each variant.

3.1 Variant A — Card Stack / List

Layout: single-column stacked cards. Navigation: standard 5-item bottom tab bar. Visual hierarchy: one hero card (AI plan) at the top of Home, with a dedicated one-line rationale directly beneath it. Feature integration: AI recommendations and social content are shown as separate, clearly bordered cards; overrides are inline icons (e.g. a swap icon next to each exercise) rather than a separate screen.

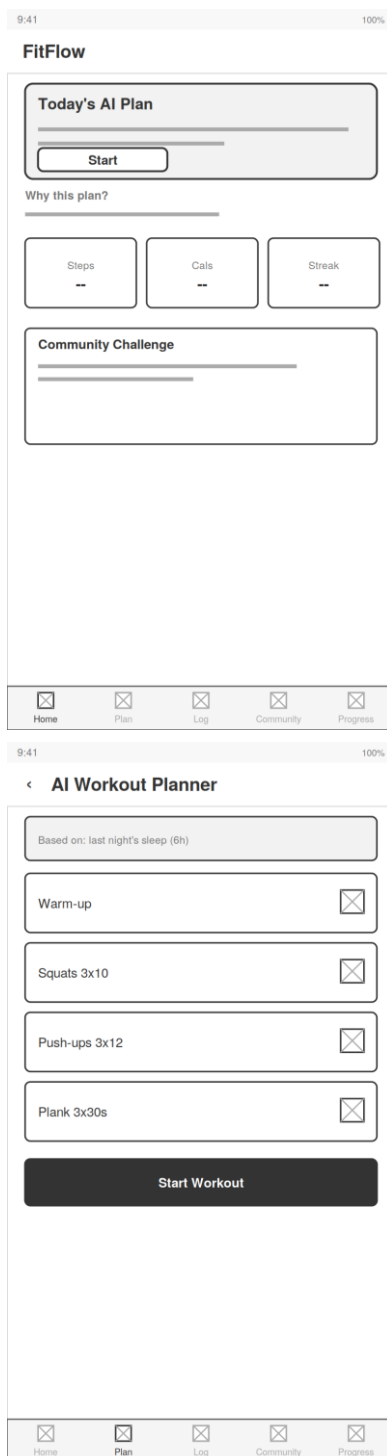


Figure 4: Variant A — Home Dashboard and AI Workout Planner.

3.2 Variant B — Segmented / Carousel

Layout: a top segmented control (Plan / Social) switches the entire Home content; the Planner becomes a single-exercise swipe able carousel instead of a list. Navigation: same bottom tab bar, but Home's segmented control adds a second layer of navigation above it. Visual hierarchy: less text density per screen

— one focal element at a time. Feature integration: overriding an exercise is a large, explicit Swap/Keep decision per card rather than a small inline icon.

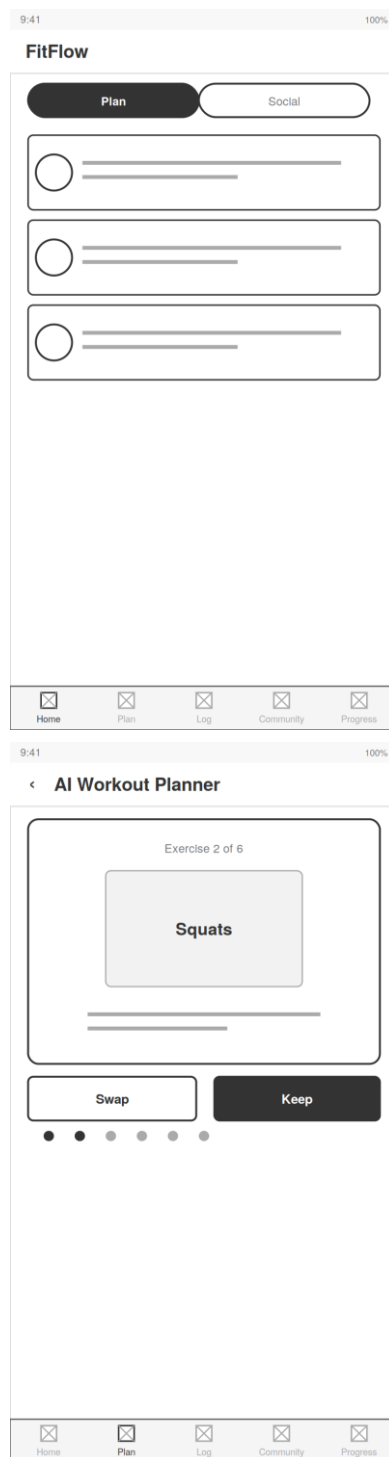


Figure 5: Variant B — Home Dashboard and AI Workout Planner.

3.3 Variant C — Progress Ring / Table

Layout: a centered progress-ring hero on Home rather than an AI-plan card; the Planner becomes a dense, spreadsheet-like table of exercises with inline editable sets/ reps. Navigation: same bottom tab bar. Visual hierarchy: strongest visual statement (large ring, high contrast CTA) but the AI rationale is one tap deeper, inside the plan screen rather than on Home. Feature integration: best suited to power users/trainers who want to scan and edit many data points at once (matches P2 and P4's requests in Lab 1).

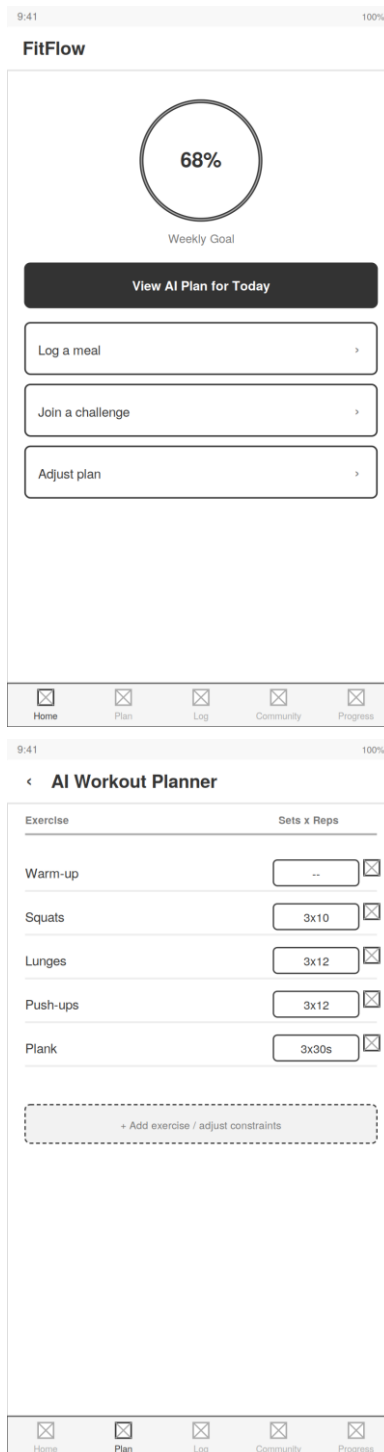


Figure 6: Variant C — Home Dashboard and AI Workout Planner.

Activity 4: Select the Best Design Using Justification

4.1 Decision Matrix

Each variant was scored 1 (poor) to 5 (excellent) against five criteria. Usability and Alignment with user needs were weighted most heavily in the final call, since they map most directly to the validated research findings from Labs 1–2.

Criterion	Variant A — Card Stack	Variant B — Segmented/Carousel	Variant C — Ring/Table
Usability	5 — familiar list pattern, low learning curve for new users (P5)	3 — segmented control adds a navigation layer; carousel hides upcoming exercises	3 — dense table is powerful but intimidating for casual/new users
Visual appeal	4 — clean, uncluttered	4 — strong focal moments, one card at a time	5 — the progress ring is the most visually distinctive moment
Alignment with user needs	5 — rationale line + inline swap icon map directly to FR-01/FR-02	3 — rationale exists but is less prominent than in A	4 — strong for power users/trainers (P2, P4) but weaker for majority-casual base
Innovation	3 — conventional, safe pattern	4 — carousel decision (Swap/Keep) is a fresh interaction	5 — ring hero and inline-editable table are the most novel
Feasibility	5 — simple components, fastest to build and test	3 — carousel state management adds engineering complexity	3 — inline table editing is the most complex to implement well
TOTAL (/25)	22	17	20

4.2 Selected Design Direction

Variant A — Card Stack / List — was selected as the primary design direction, scoring highest overall (22/25) and, more importantly, scoring the maximum on the two criteria most tied to validated research: Usability and Alignment with user needs.

4.3 Justification (approx. 350 words)

Variant A was chosen because it most directly answers the highest-priority findings from Lab 1–2 without introducing new usability risk. The single biggest theme across interviews, the survey, and usability testing was “Lack of Personalization & AI Trust” (FR-01): users such as P3 and R-27 said they didn't understand why the AI picked a given target, and trainers like P4 wanted to see the AI's reasoning before endorsing a plan for a client. Variant A puts a one-line rationale directly beneath the AI plan card on Home and repeats it on the Planner screen, so trust-building information is visible at a glance rather than buried a tap away, as it is in Variant C, or attached to a large decision moment, as in Variant B.

The second highest-priority theme was “Nutrition Tracking Issues” (FR-03), where the core complaint was raw friction — too many taps, too much searching. Variant A's simple bordered-list pattern, reused consistently for the Nutrition Logger's recent/favorite items, keeps that flow to the fewest possible taps, whereas Variant C's denser table styling and Variant B's carousel pattern both add visual or interaction overhead that would work against a friction-reduction goal.

Variant A also scored highest on feasibility, which matters because the case study's own technology stack (React Native, on-device TensorFlow Lite personalization) favors predictable, well-supported UI patterns over the more state-heavy carousel or inline-table editing patterns in Variants B and C — both of which would add engineering risk without a corresponding research-backed benefit for the majority of users.

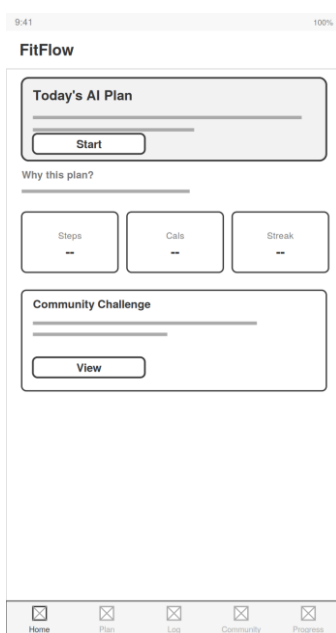
Variant A's main trade-off is that it is the least visually distinctive of the three and offers a shallower experience for power users and trainers (P2, P4) who explicitly asked for denser analytics and inline editing — a need Variant C serves better. This is a deliberate prioritization: three of five research participants and the majority of survey respondents represent the broader “casual/new user” segment for whom simplicity

and trust cues matter more than density. The recommendation is to ship Variant A as the primary experience and revisit an “advanced/trainer view” toggle — borrowing Variant C's table pattern — as a later, opt-in enhancement rather than the default.

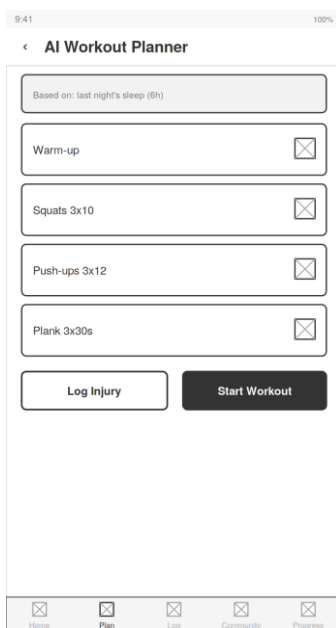
Activity 5: Paper and Digital Wireframes

Paper wireframes were hand-sketched first for all five core flows, directly extending the strongest Crazy 8 ideas (Home #1/#5, Planner #1, and equivalent list-based sketches for Nutrition Logger and Community Feed) into full-screen layouts following the selected Variant A direction. The best sketch for each flow was then digitized below as a clean, grayscale, low-fidelity wireframe using basic shapes only — no color and no detailed icons, per the deliverable requirement — while keeping the same structure, spacing, and component choices as the paper originals so nothing was reinterpreted in translation.

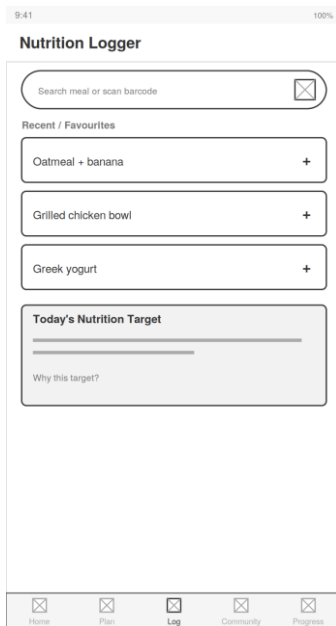
5.1 Digital Low-Fidelity Wireframes (Variant A, all 5 core flows)



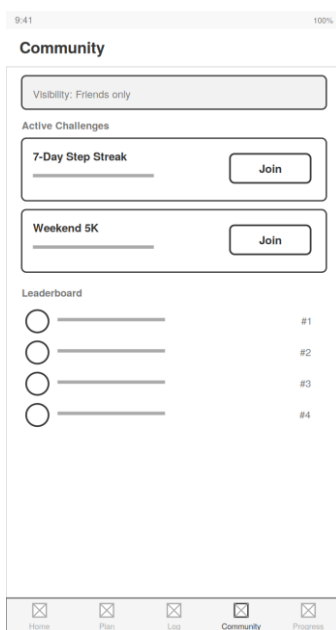
Home Dashboard



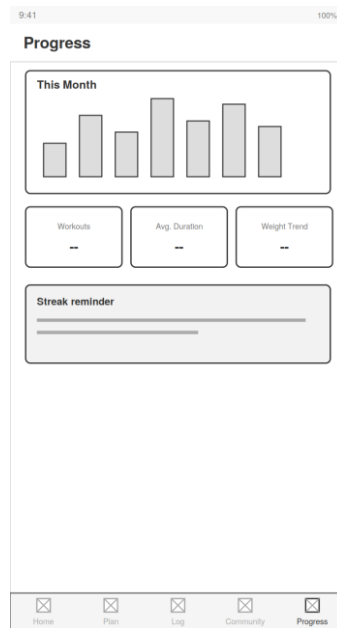
AI Workout Planner



Nutrition Logger



Community Feed



Progress Tracking

Activity 6: Clickable Low-Fidelity Prototype

All five digital wireframes were linked into a clickable, shareable low-fidelity prototype (delivered as a standalone HTML file, [FitFlow_Prototype/index.html](#), alongside this report — open it directly in any browser; no install or account needed). It mirrors the navigation structure that would be built in Figma using the same five screens and the same links, and keeps the identical grayscale, basic-shapes wireframe style used in Activity 5.

6.1 Linked Navigation Flow

FitFlow Prototype – Linked Navigation Flow (Activity 6)

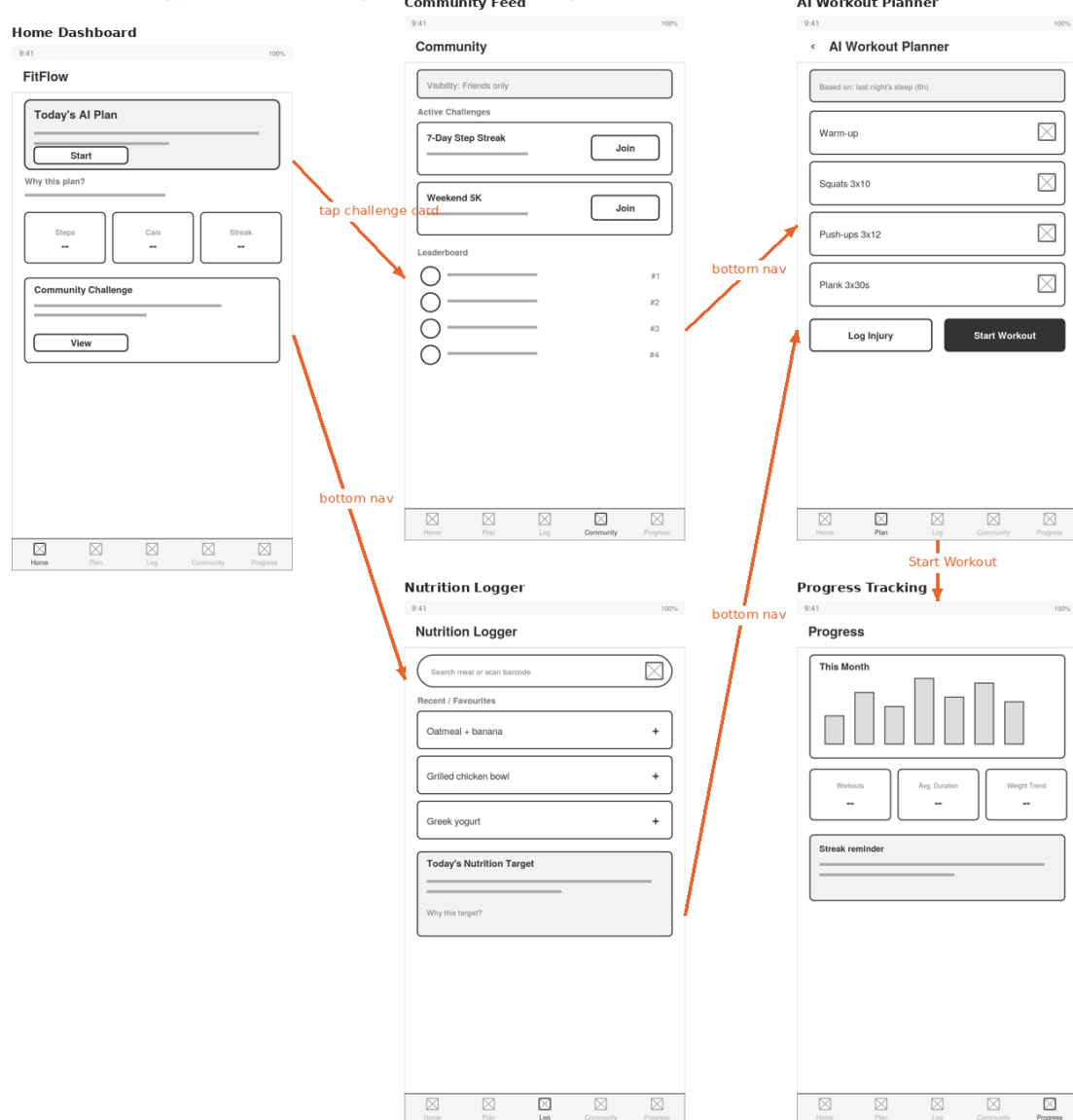


Figure 7: Linked navigation flow implemented in the clickable prototype.

6.2 Interactions Implemented

From	Trigger	To
Home	Bottom nav (Home / Plan / Log / Community / Progress)	Any of the 5 screens
Home	Tap "Start" on the AI Plan card	AI Workout Planner
Home	Tap the "Community Challenge" card	Community Feed
AI Workout Planner	Tap "Start Workout"	Progress Tracking
Any screen	Bottom nav tab	Direct jump to that screen

6.3 Annotations — Key Design Decisions

Eight in-prototype annotation markers (visible as numbered orange dots in the live file, viewable on hover/tap) document the reasoning behind specific elements, each tied back to research finding or requirement:

#	Screen / element	Design decision & rationale
1	Home – AI Plan card	Placed as the hero element so the AI plan is the first thing seen — supports FR-01/US-01.
2	Home – “Why this plan?” line	Explicit rationale text addresses the top research theme, “Lack of Personalization & AI Trust.”
3	Planner – rationale banner	Repeated (not just on Home) because trainers (P4) said they need to see the reason before endorsing a plan.
4	Planner – swap icon	Inline override control answers FR-02, requested by P2 (power user) and P4 (trainer).
5	Nutrition – search/scan bar	Dual entry (search + barcode) targets FR-03/FR-04, the top cited friction point (D-001, D-009).
6	Nutrition – one-tap “+”	Logs a recent item in a single tap, meeting the “3 taps or fewer” target in FR-03.
7	Community – visibility selector	Shown before the challenge list, directly preventing the accidental-public-join issue P5 experienced (FR-05/US-07/US-12).
8	Progress – streak reminder card	Addresses the “Poor Motivation” theme and FR-06/US-09 (proactive re-engagement).

Note: this prototype is a self-contained HTML/CSS/JS simulation of a Figma-style clickable prototype, built so it can be opened and marked without any external account. If a native Figma file is required for submission, the same five screens and links documented above can be rebuilt directly in Figma using this document as the annotated specification.

Submission Checklist

- Activity 1: Pain-point/evidence table, feature needs, prioritized requirements, and 5 core user flows (Section 1).
- Activity 2: Three Crazy 8 sketch sheets — 24 sketches total (Section 2).
- Activity 3: Three distinct interface variants across Home Dashboard and AI Workout Planner (Section 3).
- Activity 4: Decision matrix and 350-word justification for the selected design (Section 4).
- Activity 5: Digital low-fidelity grayscale wireframes for all 5 core screens (Section 5).
- Activity 6: Clickable low-fidelity prototype file, linked-flow diagram, and 8 design-decision annotations (Section 6).